

Pratyush Sudhakar

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science (*Minors: Mathematics & Psychology*)

December 2026

Coursework: Analysis of Algorithms, Systems Programming, Computer System Organization, OOP & Data Structures (Java), Adv. ML Systems (grad), Machine Learning, Computer Vision

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, OCaml, Bash/Shell

Frameworks & Libraries: React, Next.js, Node.js/Express, FastAPI, Flask, React Native (Expo), REST APIs, LangChain, PyTorch, Scikit-learn, OpenCV, NumPy/Pandas

Systems & Tools: Linux/Unix, multithreading, Docker, Kubernetes, Nginx, Git, GitHub Actions, GCP, AWS, Azure, PostgreSQL, MongoDB, Redis, accessibility (WCAG), i18n

EXPERIENCE

Full-Stack Developer, DECA Lab, Cornell University

February 2026 - Present

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, TypeScript, React Native, Next.js

Ithaca, NY

- Built two field-audit products (COPA and YEE) on web and mobile, designing typed REST APIs over a normalized PostgreSQL schema with multi-role access control, draft-save, submission, and reporting.
- Kept auditors from losing work on unreliable field networks with an offline-first React Native client (background sync), hardening draft-save against duplicate-key failures across multi-round field testing.
- Shipped accessibility and internationalization configurations for field workers with a wide range of literacy and language needs, and instrumented both products with PostHog, Sentry, and Clarity for faster bug triage.

Data Engineer, Cornell College of Engineering

June 2023 - September 2023

Tech: Python, Multithreading, Selenium, NLP, Flask, Next.js, Server-Sent Events

Ithaca, NY

- Cut wall-clock processing time 83% on an NLP extraction pipeline over financial filings for 3,000+ companies by parallelizing an I/O-bound sequential job with multithreading.
- Replaced a polling client with a Flask and Next.js Server-Sent-Events dashboard, cutting data latency 66%.

Software Engineering Intern, rapStudy (EdTech)

June 2023 - August 2023

Tech: JavaScript, React.js, Redux, Firebase, Firestore

Los Angeles, CA

- Refactored the client to Redux with memoized selectors, improving rendering speed 93% and removing the interaction lag users were dropping off on.
- Synchronized a lyric view to audio playback at 95% accuracy across variable network conditions, and opened content sharing to external users behind 10+ custom Firestore security rules.

ACTIVITIES

Head of Engineering & President, Perfect Match: Online Matchmaking

January 2023 - May 2025

Tech: Next.js, TypeScript, Flask, MongoDB, Google OR-Tools, GCP

Ithaca, NY

- Replaced a nightly batch job with a real-time Gale-Shapley stable-matching engine (Google OR-Tools), delivering 50,000+ matches to 5,000+ users with a provable no-better-pairing guarantee.
- Built a Flask analytics microservice on MongoDB Map-Reduce with caching for sub-second reads at 5,000-user scale.

Technical Product Manager, Cornell Digital Tech & Innovations

January 2023 - December 2023

Tech: React.js, DynamoDB, Cypress, Mocha, Docker, Kubernetes, GitHub Actions

Ithaca, NY

- Led 7 engineers on Carriage, the campus accessible-transportation app students with disabilities depend on, holding test coverage at 94% with Cypress and Mocha/Supertest and cutting DynamoDB operations 90%.

VENTURES & PERSONAL PROJECTS

Personal Agent Homepage: AI Dev Tooling | Python, LiteLLM, Docker, Grafana, AWS

April 2026

Runs my own coding workflow at controlled cost: each task defaults to the cheapest capable model across 8 tiers and 12+ providers, with context-window exhaustion routed separately from provider failure.

DTI GPT: RAG Assistant | Flask, LangChain, Vector DB, Azure

February 2024 - April 2024

Served natural-language queries over Cornell course syllabi at 50ms average latency from a retrieval-augmented (RAG) pipeline, iterating on chunking and reranking because vector similarity and real relevance diverge.